Jeremy Kwok

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EDUCATION

Georgia Institute of Technology - Bachelor of Science in Mechanical Engineering - *GPA: 3.7/4.0* Graduating December 2026

- **Concentration**: Micro- and Nano-Engineering, **Minor**: Computer Science (Computing and Intelligence)

Coursework: System Dynamics, Heat Transfer, Manufacturing & Design, Fluid Mechanics, Deformable Bodies, Machine Learning
Certificates: Certified SolidWorks Associate (CSWA), Lean Six Sigma Yellow Belt, Google Advanced Data Analytics

ENGINEERING EXPERIENCE

Bloom Energy (Newark, DE) - Repair & Overhaul / Engineering Intern Jun 2026 - Aug 2026

IRIS — Integrated Reliability Imaging System

- Designed and engineered Bloom's first machine vision inspection system for returned fuel cell columns, capturing burns and field damage before teardown, enabling computer vision model training and data-driven operating decisions that extend unit lifespan.
- Derived the optical specification from a 0.1 mm feature-resolution requirement to 53 µm/px across the full column face (5 MP sensor, 35 mm lens, ±13 mm depth-of-field standoff window); verified zero-gap five-camera coverage by script and rejected motorized single-camera alternative on cycle time and drift.
- Owned optical requirements, mechanical design, capture software, procurement, V&V, and handoff to operations; concept to handoff in 8 weeks, with no change to the existing teardown workflow and no contact with production tooling.
- Built automated focus and exposure acceptance gates into the capture software (~2k LOC; Python, FastAPI), with operator overrides logged to per-unit metadata so every accepted image carries its own acceptance evidence.
- Delivered a seven-document operations handoff: commissioning runbook, design decision log, V&V record with open items tracked to closure, code guide for non-programmers, and asset inventory.

Mujin (Atlanta, GA) - Product Development Intern Jan 2026 - Apr 2026

EOAT Vacuum Cup Validation • Fleet Reliability Monitoring

- Authored a two-phase supplier-change validation plan qualifying a replacement vacuum cup on a 63-cup depalletizing gripper, with adopt / reject / limited-deployment decision logic, must-not-worsen acceptance criteria, and metrics per 10,000 picks.
- Ran Phase 1 bench testing on a force/vacuum jig with KEYENCE digital pressure sensing: pull-off force, 10-second static retention, and vacuum decay across a 2–10 lb load range and five surface conditions, against a ≤10% decay acceptance gate.
- Structured Phase 2 as a 30/60/90-day A/B field comparison on a live production line, stratified by weight and surface type, with a failure mode framework supporting the go/no-go decision; led field testing at a 24/7 customer distribution center.
- The plan's field phase caught a durability failure that bench testing missed: replacement cups cracked and wore within about two weeks, and the pre-set criteria called a no-swap before any fleet-wide rollout; two new metrics were added to fleet monitoring.
- Built StatsMonitor (~5.8k LOC Python), a fleet reliability pipeline covering 13 controllers at 6 customer sites whose evidence-gated daily KPI report and ranked watchlist the service team uses to decide which units get attention first.

BIOMEDICAL RESEARCH

Sulchek Lab, Georgia Tech (Atlanta, GA) - Undergraduate Researcher Jun 2025 - Present

AFM Measurement & Damage Characterization

- Owned the damage analysis for a 397-indentation AFM dataset (102 naive T cells across four post-thaw days), with report numbers regenerated from scripts and a regression baseline that must reproduce after every change.
- Showed that achieved (not commanded) indentation load and rate account for several times more modulus variance than measurement day; built the harmonization pipeline to carry achieved values as covariates rather than assume setpoints.
- Built cell-detection pipeline (85-frame series averaging, fixed-pattern dust subtraction, and per-session cantilever drift registration lifting one day's target detection from 62% to 96%), delivered 31,000+ target/bystander cell crops to lab's downstream pipeline.

PROJECTS & LEADERSHIP

BlockParty - Autonomous Competition Robot (Best Design Award) Jan 2025 - May 2025

- Over three judged sprints, designed, fabricated, and tested mechanisms for a fully autonomous robot (pneumatic pistons, solenoids, motors, mouse-trap actuators, Arduino control), with concepts chosen by morph chart and weighted evaluation matrix.

Enport - Engineering Portfolio Platform - *Founder* Jan 2026 - Oct 2026

- Built a Next.js/TypeScript platform that presents engineering projects as structured claims tied to evidence, with schema validation and publish checks; it now powers my portfolio at jeremy-kwok.github.io.

FRC Robotics: Excelsior and Jim - *Director of Pneumatics* Jan 2022 - Jun 2023

- CAD'd and built the swerve drivetrain and one-piece bumpers (metal chassis cut and filed, 8 swerve modules assembled and tested, bellyplate riveted); designed and maintained two pneumatic systems across two different robots for competition seasons.

Develop For Good (Remote: Stanford, CA) - *Product Lead (prev. Engineering & Product Manager)* Apr 2023 - Dec 2025

- Directed 9 project teams building tech solutions for nonprofit clients; trained 150+ managers in product management, assessed 600+ program applications, and revised 40+ Product Requirements Documents, roadmaps, and engineering design documents.

SKILLS

- **CAD & Analysis:** SolidWorks (CSWA), Onshape, MATLAB, OpenFOAM / ParaView, Excel, Machine Vision Optics
- **Fabrication & Test:** Machine Shop Tools, Soldering, 3D Printing, Laser Cutting, Pneumatics, Digital Pressure Instrument
- **Programming & Statistics:** Python (NumPy, pandas, scikit-learn), C++, Java, JavaScript, Git / GitHub, Jupyter / Colab